

Cauda equina syndrome (CES) investigation

Use in all adults presenting with leg pain and/or back pain AND EITHER new onset (within last 14 days) OR a deterioration of any of the features listed in Box 1

DO NOT use if patient has

- Known malignancy capable of spreading to the spine (follow UHL MSCC guideline)
- Significantly abnormal vital signs (seek systemic/visceral causes)
- Been referred to T&O team by GP (ortho team to manage)

Disclaimer: This is a clinical template; clinicians should always use judgment when managing individual patients

Re-approved by ED guidelines committee on 09Apr25
Review due Apr 2030 . Trust Ref: C53/2021

Patient details

Full name

DoB

Unit number

(use sticker if available)

① CES investigation indicated?

☐ **Yes** One or more of the red flags below

- Sense of numbness or pins & needles around anus, genitals or inner thighs ☐
- Difficulty initiating micturition ☐
- Impaired sensation of urinary flow ☐
- Painless urinary retention ☐
- Urinary incontinence (**unless** features suggest stress or urge incontinence) ☐
- Severe or progressive bilateral leg weakness affecting knee extension, ankle eversion or foot dorsiflexion ☐
- Loss of sensation of rectal fullness ☐
- Faecal incontinence ☐
- Inability to achieve erection or ejaculate ☐
- Loss of genital sensation ☐
- Bilateral radicular (below knee) leg pain of sudden onset, or sciatica that has progressed from unilateral to bilateral ☐

☐ **NO** None of the red flags above. MRI not indicated. File pathway in pt's notes.

② Analgesia bundle

NB: Account for analgesia taken prior to arrival before prescribing

in NC Meds, go to Emergency Medicine (ED) > Analgesia (ED) > Systemic analgesia and select:

- Paracetamol PO
- Dihydrocodeine PO if not drowsy; use codeine instead if aged >65
- Diclofenac 100mg PR if safe (**NB:** If aged >65 prescribe ibuprofen instead)
- If pain score > 6/10, oramorph **OR** (rarely) morphine IV
- DO NOT** prescribe benzodiazepines

③ NSAID safety issues?

☐ **Yes**, at least one of the below

- Patient allergic to any NSAID ☐
- Exacerbation of asthma after use of any NSAID in the past ☐
- Known current peptic ulcer ☐
- Known or obvious heart failure ☐
- Known inflammatory bowel disease ☐
- Currently treated with aspirin ☐
- Known abnormal renal function ☐
- Current illness with risk of AKI ☐

☐ **NO**, none of the above

- Meticulously document onset date (and time if within last 48h) of each red flag
 - Send FBC, U&E, LFT and CRP
 - Provide effective analgesia (see boxes 2 and 3)
 - Perform neurological leg and perianal sensation exam; record findings in box 7
 - Ask patient to void and arrange bladder scan within 5min; record PVR in box 4
- NB:** Tick if unable to obtain PVR because ☐ catheter in situ or ☐ in retention

Manage as appropriate

Any reason to admit?

☐ Y ☐ N

CES-I

CES incomplete; better recovery chance with rapid decompression

CES-R

CES with urinary retention; chance of recovery even with rapid surgical intervention poor

If in AUR, complete box 5

Admit to EDU on 'CES investigation pathway' while MRI awaited (**NB:** Not suitable for EDU if abnormal bloods or vital signs, or other reasons to admit; keep in ED instead)

* Request MRI L-spine (**NB:** Ask area lead/consultant/EPIC to do)

On the request, record information below in the 'Clinical question to be answered' box:

- Cauda equina compression?
- Other structural cause of pain?
- Non-compressive pathology?

Add relevant clinical information including PVR (or retention)

Between 20:00 and 08:00, MRI request should state if CES-I suspected and scan urgent or if wait until 07:30 is acceptable

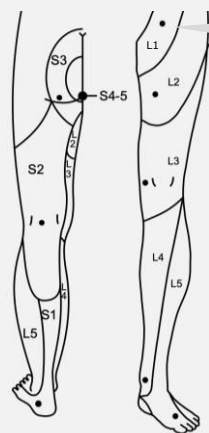
Vetting of MRI requests

- 08:00 - 19:30 call not needed
- 19:30 - 08:00 call 16969

MRI radiographer extensions 17746 and 17747

Warning signs: CES may occur
Urgent (two-week wait) review by spinal team indicated

Discharge with 'At risk of CES' bundle (complete box 6)

⑦ Focused neurological examination

Sensation is best tested in the places marked by dots

Motor exam findings

- Knee extensors
- Ankle dorsiflexors
- Long toe extensors
- Ankle plantar flexors

Deep tendon reflexes

- Patellar
- Ankle

R L

Motor power grading

- 0 = Total paralysis
- 1 = Palpable or visible contraction only
- 2 = Active movement, gravity eliminated
- 3 = Active movement, against gravity
- 4 = Active movement, against some resistance
- 5 = Active movement, against full resistance

NT = Not testable

0*, 1*, 2*, 3*, 4* = Non-SCI condition present

Sensation exam findings

	M	LT	PP	LT	PP	M
L1						
L2						
L3						
L4						
L5						
S1						
S2						
S3						
S4-5						
Right leg						
Left leg						

Reflex grading

- 0 = no response; abnormal
- 1+ = slight but definite response; may or may not be normal
- 2+ = brisk response; normal
- 3+ = very brisk response; may or may not be normal
- 4+ = tap elicits a repeating reflex (clonus); abnormal

④ Post-void residual (PVR)

NB: Complete bladder scan within 5min of voiding

mL

⑤ AUR documentation

Record bladder volume

mL

Sensation of catheter insertion

☐ Present ☐ Absent

Sensation of catheter tug

☐ Present ☐ Absent

⑥ 'At risk of CES' bundle

- Refer to spinal team via UHL PathPoint ('eTrauma' - see box 8 for guidance) ☐
- Print out the CES warning card in patient's preferred language ☐
- Add front page of completed proforma as clinical photograph on Nervecentre ☐
- Give patient a copy of this proforma ☐
- Signpost patient to the MACP video (see box 9 for QR code with the link) ☐
- Prescribe appropriate TTO analgesia ☐
- NB:** Tick ☐ if no analgesia needed

Seen by

Print name

Signature

Role

Date

⑧ **2-week wait referral to spinal team via eTrauma 'virtual fracture clinic' (VFC)**

Log into UHL PathPoint [eTrauma](#) (through this link or via NC), and start referral. Complete mandatory fields (marked with *) as shown.

Date of injury *

10/01/2026

Select date bilateral symptoms started

Diagnosis *

Bilateral sciatica ✕

Side *

Right and left ▼

Mechanism of injury *

None ▼

Actions taken *

Advice given ✕

For the 'Referral notes' box, copy & paste text below ('Paste text only' option will preserve format). Add / delete [\[including help text\]](#) as appropriate.

Referral notes *

Styles ▼ **B** *I* U ~~T~~ | ▼ | Source ▼

I am requesting a two-week wait appointment with the spinal team due to the following warning symptoms that CES may occur [[delete the non-applicable option](#)]:

- Sudden onset bilateral radicular pain without CES symptoms
- Unilateral radicular leg pain that has progressed to bilateral without CES symptoms

[[insert a brief summary of symptoms - including duration and progression - and examination findings](#)]

A clinical photograph of the completed ED CES investigation proforma is available on Nervecentre.

We have provided the patient with

- A copy of the completed ED CES investigation proforma
- A copy of the CES warning card
- A QR code of the MACP CES warning sign video link

Patient guidance

- None -

Keep as 'None' (no specific UHL PIL available)

Save VFC trauma referral

Referral will be sent on clicking this button

⑨ **CES warning signs – information for patients**

Scan the QR code to watch the three-minute [video](#) **'When you should seek urgent help for your back pain'** by The MACP (Musculoskeletal Association of Chartered Physiotherapists)

